Value Stream Mapping Report

Process Name: Custom Widget Manufacturing Process

Customer: ABC Manufacturing Corp

Supplier: XYZ Components Ltd

Product: Custom Industrial Widget

Analysis Date: 2025-07-31

Analyst: VSM Generator System

Key Metrics Summary

Total Cycle Time:	137.0 minutes
Total Lead Time:	2175.0 minutes
Value-Added Time:	120.0 minutes
Process Efficiency:	87.6%
Takt Time:	30.0 minutes
Number of Processes:	7

Executive Summary

This value stream analysis examines the Custom Widget Manufacturing Process process serving customer ABC Manufacturing Corp. The analysis identified 7 key process steps with a total cycle time of 137.0 minutes and lead time of 2175.0 minutes. The current process efficiency is 87.6%, indicating significant opportunities for improvement through waste elimination and process optimization.

Waste Analysis Summary:
Total identified waste: 5496.0 minutes
Primary waste sources: transportation, inventory, motion, waiting, overprocessing
Improvement potential: 252.7% lead time reduction

Process Details

Process	Cycle Time	Lead Time	Uptime	Operators	Efficiency	Value Added
Order Processing	15.0 min	120.0 min	95%	2	12.5%	Yes
Material Preparation	30.0 min	240.0 min	85%	3	12.5%	Yes
Quality Check	10.0 min	60.0 min	98%	1	16.7%	Yes
Assembly	45.0 min	180.0 min	90%	4	25.0%	Yes
Final Inspection	20.0 min	90.0 min	95%	2	22.2%	Yes
Packaging	12.0 min	45.0 min	92%	2	26.7%	No
Shipping	5.0 min	1440.0 min	80%	1	0.3%	No

Waste Analysis

Waste Type	Impact (minutes)	Description
Waiting	2038.0	Idle time waiting for materials, information, or equipment
Transportation	1645.0	Unnecessary movement of materials or information
Overprocessing	1468.0	More work than required by customer
Inventory	210.0	Excess raw materials, WIP, or finished goods
Motion	135.0	Unnecessary movement of people

Improvement Recommendations

Recommendation 1: Flow Balance

Priority: High
Description: Balance line to takt time (30.0 min)
Action: Add capacity to bottleneck processes or redistribute work
Expected Impact: Improve flow and reduce waiting

Recommendation 2: Waste Elimination

Priority: Medium
Description: Reduce transportation waste
Action: Unnecessary movement of materials or information
Expected Impact: Potential 1645.0 min reduction in lead time

Recommendation 3: Waste Elimination

Priority: Medium
Description: Reduce inventory waste
Action: Excess raw materials, WIP, or finished goods
Expected Impact: Potential 210.0 min reduction in lead time

Recommendation 4: Waste Elimination

Priority: Medium
Description: Reduce motion waste
Action: Unnecessary movement of people
Expected Impact: Potential 135.0 min reduction in lead time

Recommendation 5: Waste Elimination

Priority: Medium
Description: Reduce waiting waste
Action: Idle time waiting for materials, information, or equipment
Expected Impact: Potential 2038.0 min reduction in lead time

Recommendation 6: Waste Elimination

Priority: Medium
Description: Reduce overprocessing waste
Action: More work than required by customer
Expected Impact: Potential 1468.0 min reduction in lead time

Recommendation 7: Equipment Reliability

Priority: Medium
Description: Improve equipment uptime
Action: Implement TPM (Total Productive Maintenance)
Expected Impact: Increase overall capacity and reduce variability